

Agentic I2ICF for SDV in C-ITS

AI Agent-based Orchestration
of In-Network Computing Functions

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1. Background and Motivation

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Software-Defined Vehicles

- Vehicle functions are increasingly implemented as **software services**.
- Perception, planning, control, diagnostics, and infotainment are **continuously updated**.
- SDVs require **flexible coordination** with roadside, mobile, and cloud infrastructure.

2



In-Network Computing

- Programmable network devices can process data **inside the network**.
- ICFs can support **aggregation, filtering, security, analytics, and edge offloading**.
- However, there is no **unified framework** for registering, configuring, managing, and monitoring ICFs.

3



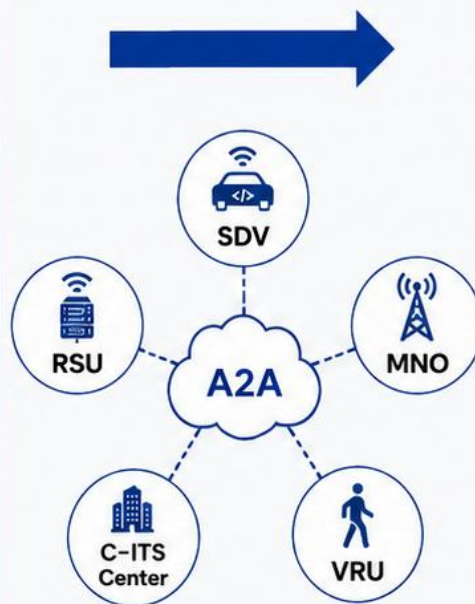
Need for Agentic Coordination

- C-ITS consists of **heterogeneous entities** across multiple domains.
- Static interfaces and message-based control are **insufficient for dynamic SDV services**.
- **AI agents and A2A communication** can support capability discovery, resource negotiation, task delegation, and adaptive ICF placement.

2. Problem and Key Idea

! Problem

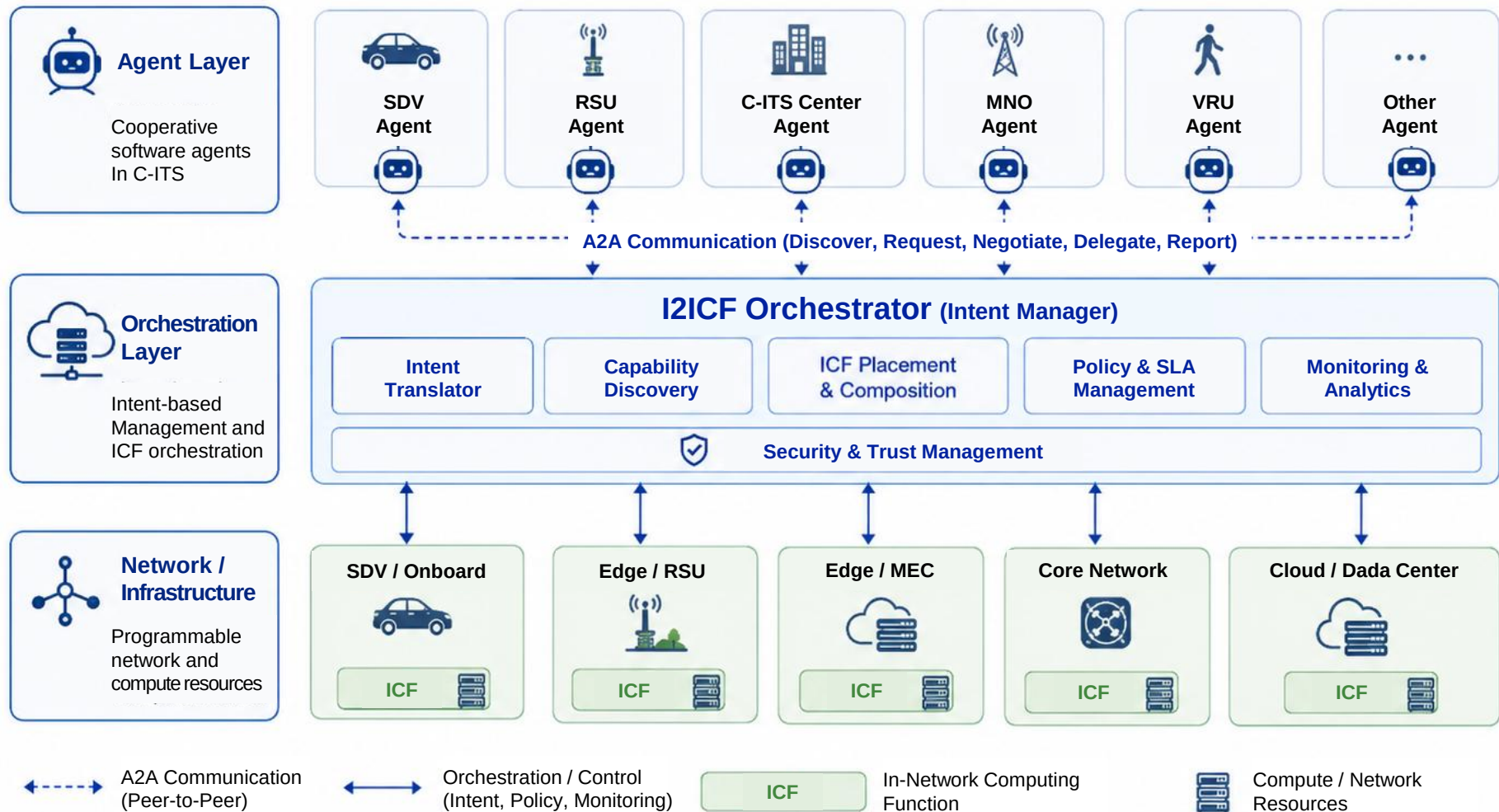
- **No structured framework** to orchestrate, manage, and monitor ICFs in SDV-centric C-ITS.
- C-ITS entities are **heterogeneous and distributed** across vehicles, roadside infrastructure, and mobile networks.
- **Static interfaces are insufficient** for dynamic service coordination and real-time decision-making.
- **Multi-domain resources** and edge functions must be selected and coordinated in a timely manner.



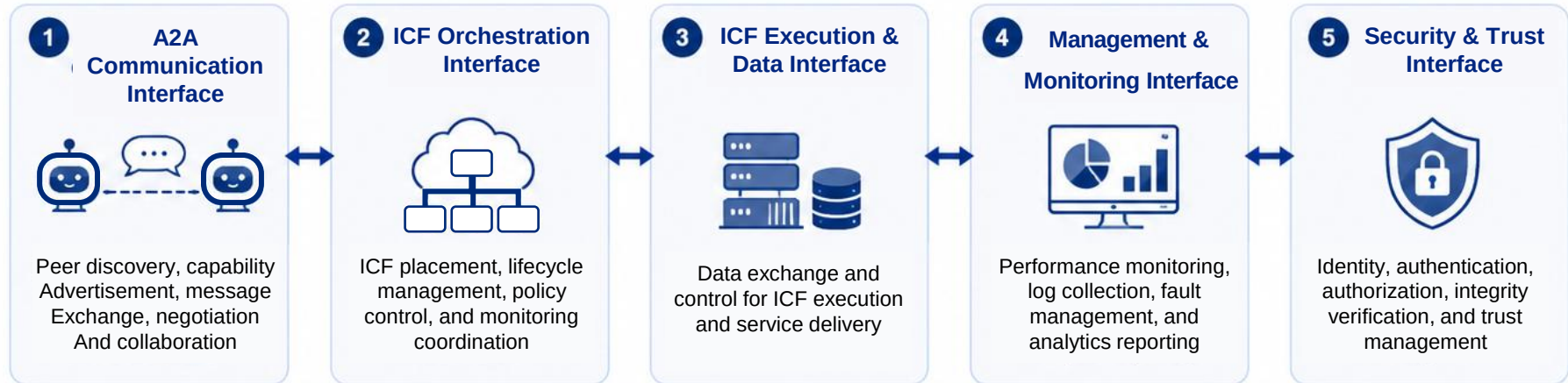
💡 Key Idea

- Deploy an **AI Agent with each functional entity** such as SDV, RSU, C-ITS Center, MNO, and VRU.
- Use an **A2A communication overlay** for capability advertisement, peer discovery, and resource negotiation.
- **Coordinate ICF placement and execution** across multi-domain networks based on context and service needs.
- Enable **efficient V2V/V2I support, edge data processing, and adaptive service orchestration**.

3. Architecture

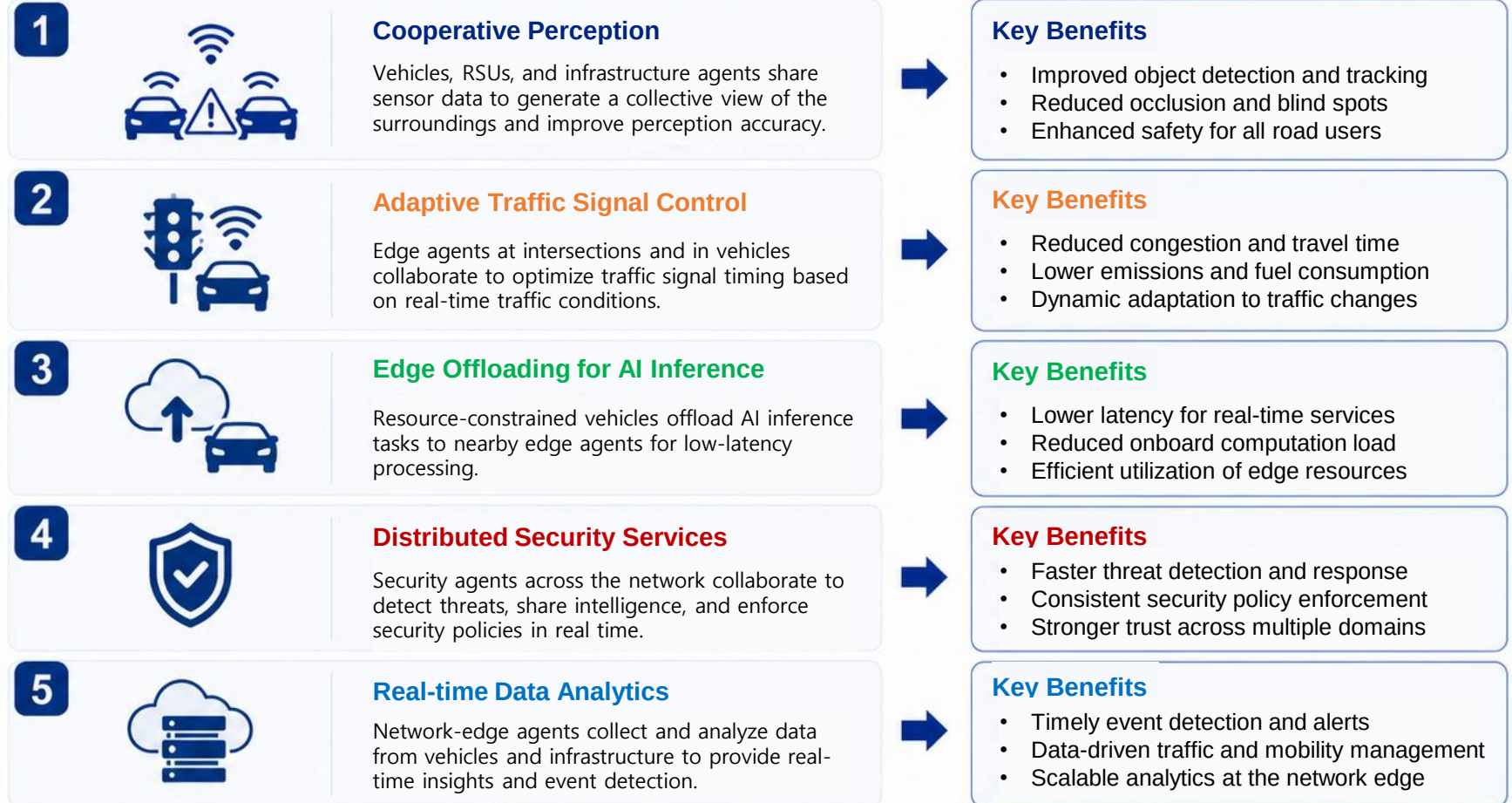


4. Interfaces



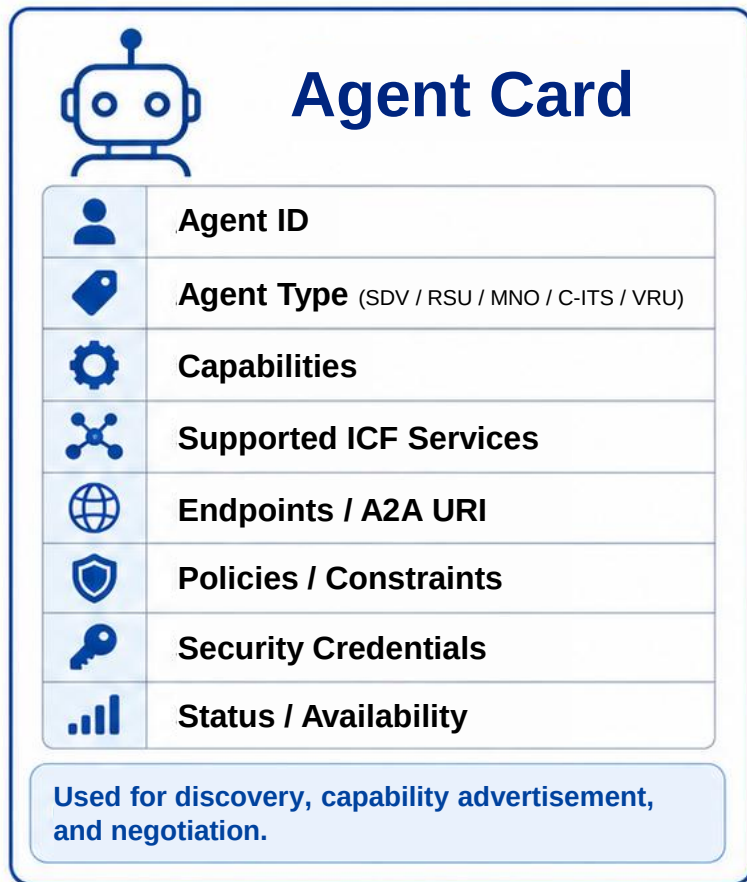
Interface	Purpose	Main Functions	Related Standards / Technologies
1 A2A Communication Interface	Enable AI agents to discover, communicate, and collaborate	Capability advertisement, peer discovery, message exchange, negotiation, delegation	IETF A2A, HTTP/2, JSON/CBOR, OAuth 2.0, SCID, DID
2 ICF Orchestration Interface	Orchestrate ICF lifecycle and resource allocation	ICF placement, instantiation, scaling, update, termination, policy control	IETF YANG (I2ICF), RESTCONF, NETCONF, Intent APIs
3 ICF Execution & Data Interface	Exchange data and control ICF execution	Data streaming, event notification, service invocation, result collection	gRPC, MQTT, WebSocket, QUIC, YANG Push
4 Management & Monitoring Interface	Monitor and manage ICF and system performance	Metrics collection, logging, alarm, fault management, analytics	Prometheus, OpenTelemetry, Syslog, NETCONF/YANG
5 Security & Trust Interface	Provide secure and trusted agent interactions	Authentication, authorization, attestation, integrity, trust evaluation	TLS 1.3, OAuth 2.0, SCITT, TPM, IETF RATS, YANG Security Models

5. Use Cases



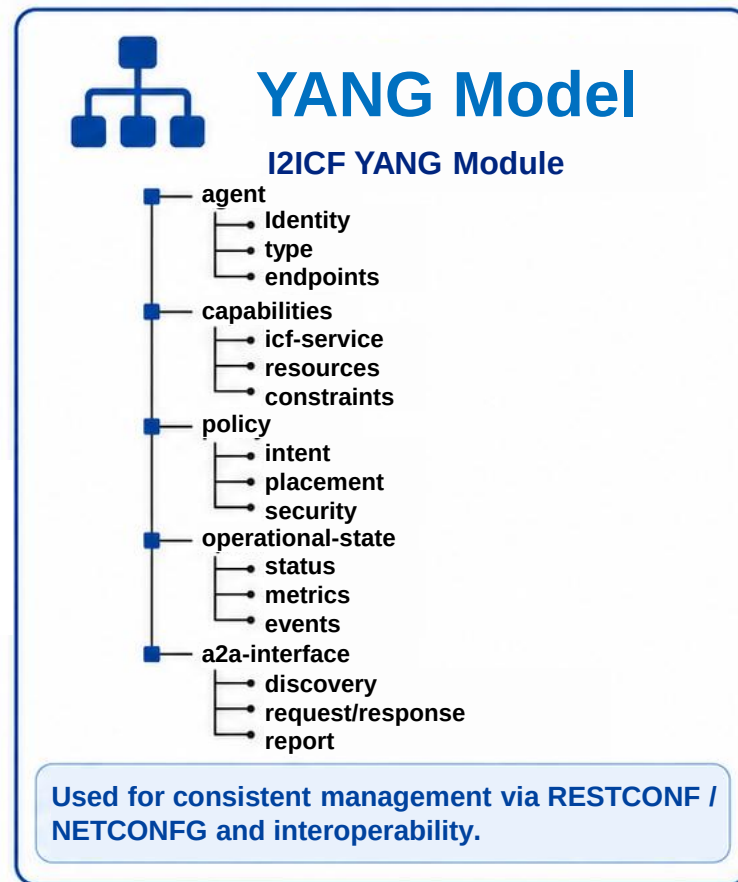
6. Agent Card / YANG Model

- ❖ **Agent Card** describes what an agent is, what it can do, and how it can be reached.
- ❖ **YANG model** provides a structured data model for configuration, orchestration, and monitoring



➔

From agent
description
to structured
management



7. Operational Considerations

- ❖ Deploying and operating Agentic I2ICF in C-ITS and SDV environments requires careful consideration of **scalability**, **reliability**, **security**, **interoperability**, and **performance**.
- ❖ These considerations ensure **robust and efficient in-network computing services** across diverse domains and use cases.

1 Scalability



- Support dynamic scaling of ICF instances based on demand.
- Handle massive numbers of devices, agents, and services.
- Enable horizontal scaling across distributed infrastructure.

2 Reliability & Availability



- Ensure high availability of ICF services with redundancy and failover.
- Support fault detection, recovery, and self-healing mechanisms.
- Maintain service continuity in dynamic network conditions.

3 Security & Privacy



- Secure agent communications, including A2A, orchestration, and data exchange.
- Protect data privacy and comply with regulations.
- Support identity, authentication, and authorization management.

4 Interoperability



- Ensure interoperability across heterogeneous systems, networks, and administrative domains.
- Use open standards and YANG models for consistent integration.
- Enable multi-vendor and cross-domain collaboration.

5 Performance & Latency



- Meet strict latency requirements for real-time C-ITS and SDV services.
- Optimize ICF placement and resource allocation.
- Monitor and optimize end-to-end performance.

6 Observability & Monitoring



- Provide real-time monitoring of agents, ICFs, and network resources.
- Collect metrics, logs, and events for analysis.
- Enable SLA monitoring and proactive issue detection.

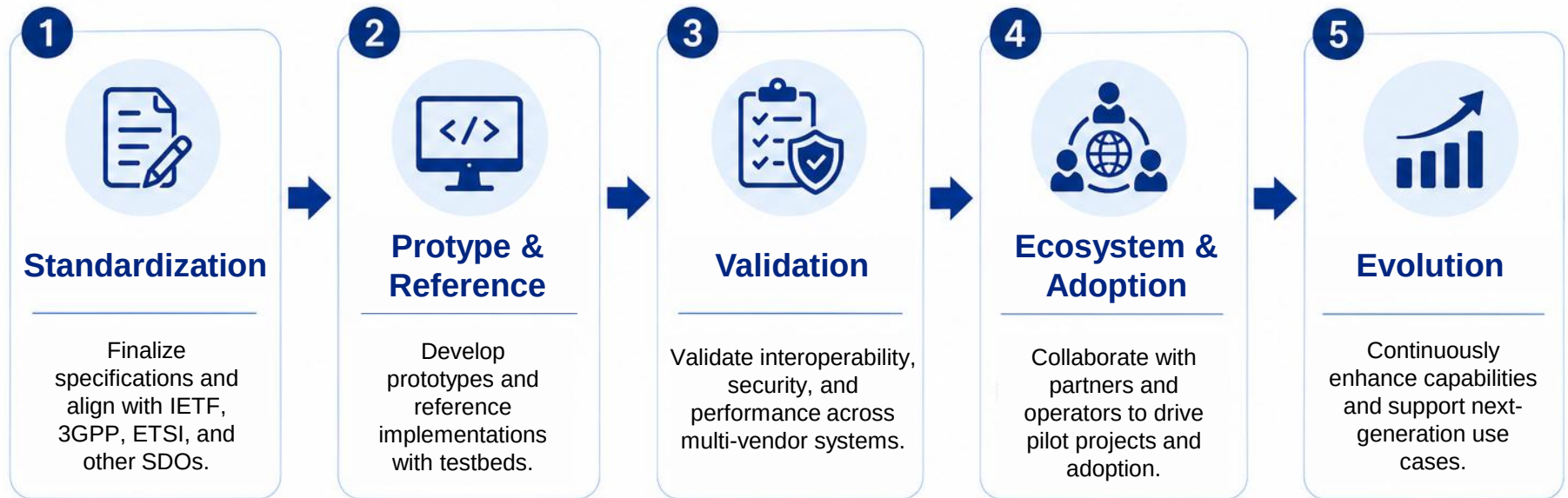
7 Automation & Lifecycle Management



- Automate the ICF lifecycle, including deployment, update, scaling, healing, and termination.
- Support intent-driven automation and policy enforcement.
- Reduce operational complexity and human intervention.

8. Next Steps

- ❖ Agentic I2ICF will be advanced through **standardization**, **prototyping**, **validation**, and **ecosystem building** to drive adoption in C-ITS and SDV environments.
- ❖ We will deliver **interoperable**, **secure**, and **scalable** in-network computing services together with industry and standards bodies.



Deliver interoperable, secure, and scalable in-network computing services for C-ITS and SDV.



Unframed Perspective

"틀에서 벗어난 시각으로 미래를 이끌어 간다"



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